

SATQUERY AI — EARTH OBSERVATION REPORT

Remote Sensing Inspection & Telemetry

Report ID:	3f554cad8de749d3	Generated:	2026-09-07T18:39:09
Confidence:	High (0.96)	Imagery Count:	1 scene(s)
Visual Query:	Analyze agricultural crop health and identify the river channel	Evidence Type:	bbox

Executive Summary & Findings

Inland lacustrine feature localized in the southern/western quadrant with stable hydrologic perimeter boundaries.

Visual Evidence Grounding



Multi-Angle Satellite Telemetry

Scene Overview	Hydrological drainage basin and shoreline environment displaying distinct spectral contrast against surrounding riparian terrain.
Land Cover (LULC)	Open Water: ~45%, Riparian Vegetation: ~35%, Adjacent Open Ground: ~20%
Key Features	- Principal water basin situated in the lower-left to central sector - Shoreline transitional buffer zone with characteristic low NIR albedo
Spatial Patterns	Naturally meandering shoreline geometry consistent with alluvial deposition patterns.
Spectral Indicators	Deep optical absorption confirming active surface water; marginal turbidity along shallow embankments.
Risk & Exposure	Monitoring recommended for shoreline erosion and seasonal water-level fluctuation.

Execution Trace & Telemetry Audit

Stage	Status	Detail	Latency
Compatibility Check	OK	Validated 1 image(s) for intent: single_image_vqa	1.0ms
Routing	OK	Intent: single_image_vqa → Selected Specialist: RS-VLM	0.0ms
Analyzing	OK	Fine-Tuned RS-VLM (BigEarthNet.txt Qwen2-VL Specialist)	2149.0ms
Building Evidence	OK	Rendered bounding box overlay: [10.0, 45.0, 55.0, 90.0]	415.0ms
Confidence Scoring	OK	Raw: 0.96 → High (0.96)	0.0ms